

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (CE) Examination

March-April 2018

CE411 Service Oriented Architecture

Date: 04.04.2018, Wednesday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Answer the following questions.**

- A. Justify: Service oriented architecture is loosely coupled. [01]
- B. Which of the following is commonly used to describe the service interface, and how to bind information, along with the nature of the component's service or endpoint? [01]
a) WSDL b) SCDL c) XML d) JSON
- C. What are the IT benefits of SOA? [03]
- D. "All valid XML files are well-formed" State True or False with justification. [02]
- E. What are the challenges for Service Oriented Architecture? [04]

Q – 2.A List and explain different characteristics of Services. [04]**Q – 2.B Answer the following questions [Any Two].** [08]

- A. What is message exchange pattern? Explain Request-Response and publisher and subscriber MEPs.
- B. Explain concept of distributed computing. Compare centralized computing with distributed computing.
- C. List SOA Architectural styles. Explain any four styles in details.

Q - 3 Answer the following questions. [Any Two]. [12]

- A. Explain Service Life Cycle with example.
- B. Explain term : (i) Orchestration (ii) Choreography
- C. Explain the term : (i) WS-Notification Framework (ii) WS-Addressing

SECTION – II**Q - 4 Answer the following questions.**

- A. Which of the following is used to define the service component that performs the service ? [02]
a) WSDL b) SCDL c) XML d) None of the mentioned
- B. Define Correlation in SOA. [02]

- C. Differentiate between DTD and XSD schema. [02]
- D. Design a XML Document for “University Exam Time Table.” [02]
- E. Explain Model Driven Architecture. [03]
- Q – 5.A Write XSD file for student.xml where studentId is mandatory and unique, AadharNo is mandatory, contact number should be zero or more. Also write sample student.xml file. [06]
- Q – 5.B Write WSDL Code to get maximum value among three different values. [06]

OR

- Q – 5.A Draw and explain SOA Reference Architecture. [06]
- Q – 5.B What is web service? How it differs than website. Explain each WSDL tag in details with example. [06]
- Q – 6. **Answer the following. [Attempt any Three]** [12]
 - A. Discuss J2EE Architecture with each modules
 - B. Discuss .NET Architecture with each modules
 - C. Explain top-down strategy of Service Oriented Architecture
 - D. What are Service endpoints and service descriptions? Explain each in brief.